Credit Card Customer Segmentation Using K-Means

1. Business Framing

Objective

The objective of this analysis is to build a **customer segmentation model** using credit card usage behavior in order to support **data-driven marketing strategies**. By grouping customers with similar behaviors, the business can design targeted interventions such as: - Credit limit upselling - Rewards and loyalty programs - Installment-based promotions - Risk mitigation for cash-advance-heavy users

Why K-Means Clustering?

- No labeled target variable exists
- Goal is to discover **natural behavioral groupings**
- Works well for numeric, continuous variables
- Scalable and easy to explain to business stakeholders

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2. Data Understanding

Data Dictionary

Variable Name	Description	Data Type	Typical Range / Notes	Business Interpretation
CUST_ID	Unique identifier of the credit card holder	Categorical	Unique per customer	Used for identification only; excluded from modeling
BALANCE	Outstanding balance available for purchases	Numeric (Continuous)	≥ 0	Indicates revolving balance and potential credit utilization
BALANCE_FREQUENCY	Frequency of balance updates	Numeric (Ratio)	0–1	Higher values indicate active account usage

Variable Name	Description	Data Type	Typical Range / Notes	Business Interpretation
PURCHASES	Total amount of purchases made	Numeric (Continuous)	≥ 0	Overall spending intensity
ONEOFF_PURCHASES	Total amount of one-time (lump sum) purchases	Numeric (Continuous)	≥ 0	Indicates preference for single large purchases
INSTALLMENTS_PURCHASES	Total amount of installment purchases	Numeric (Continuous)	≥ 0	Shows tendency toward EMI / installment plans
CASH_ADVANCE	Total cash advance amount taken	Numeric (Continuous)	≥ 0	Proxy for liquidity stress or short-term cash needs
PURCHASES_FREQUENCY	Frequency of purchase activity	Numeric (Ratio)	0–1	Measures how often purchases are made
ONEOFFPURCHASESFREQUENCY	Frequency of one-off purchases	Numeric (Ratio)	0–1	Distinguishes lump-sum spenders
PURCHASESINSTALLMENTSFREQUENCY	Frequency of installment purchases	Numeric (Ratio)	0–1	Indicates structured payment behavior
CASHADVANCEFREQUENCY	Frequency of cash advances	Numeric (Ratio)	0–1	Higher values indicate reliance on cash advances
CASHADVANCETRX	Number of cash advance transactions	Numeric (Discrete)	Integer ≥ 0	Measures intensity of cash advance usage

Variable Name	Description	Data Type	Typical Range / Notes	Business Interpretation
PURCHASES_TRX	Number of purchase transactions	Numeric (Discrete)	Integer ≥ 0	Captures transaction-level engagement
CREDIT_LIMIT	Credit card limit assigned to the user	Numeric (Continuous)	≥ 0	Indicates customer credit capacity
PAYMENTS	Total payments made by the user	Numeric (Continuous)	≥ 0	Reflects repayment behavior
MINIMUM_PAYMENTS	Minimum payment amount paid	Numeric (Continuous)	≥ 0 (may be missing)	Proxy for repayment discipline
PRCFULLPAYMENT	Percentage of full balance paid	Numeric (Ratio)	0–1	Higher values indicate responsible repayment
TENURE	Length of relationship with the bank (months)	Numeric (Discrete)	Typically 6–12	Indicates customer maturity

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3. Data Preprocessing

3.1 Remove Identifier

- Drop `CUST_ID` as it has no behavioral meaning and should not influence clustering.

3.2 Handling Missing Values

- Missing values are commonly observed in `MINIMUM_PAYMENTS` and `CREDIT_LIMIT`.
- Use **median imputation** to avoid distortion from extreme values.

Business justification: Median imputation ensures typical customer behavior is preserved without being influenced by outliers.

3.3 Outlier Awareness

- Variables such as BALANCE, PURCHASES, and CASH_ADVANCE are right-skewed.

- Do not remove outliers blindly, as high spenders are often valuable customers.
- Scaling will mitigate magnitude dominance.

3.4 Feature Scaling

- Apply **StandardScaler** to normalize all numeric variables.
- Scaling is mandatory since K-Means relies on Euclidean distance.

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4. Exploratory Data Analysis (EDA)

4.1 Distribution Analysis

- Spending and cash advance amounts are highly skewed.
- Majority of users are low to moderate spenders.
- A small proportion of customers drive a large share of transaction value.

Business insight: Customer behavior is heterogeneous, making segmentation essential.

4.2 Correlation Analysis

- PURCHASES correlates with PURCHASES_TRX
- CASH_ADVANCE correlates with CASHADVANCETRX
- PAYMENTS correlates with PRCFULLPAYMENT

Highly correlated features are retained if they capture different behavioral dimensions (frequency vs volume).

4.3 Behavioral Questions

EDA should help answer: - Who spends heavily but pays minimum balances? - Who relies on cash advances? - Who prefers installment purchases?

These questions guide feature engineering.

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5. Feature Engineering

5.1 Derived Behavioral Features (Optional)

- Installment Share = INSTALLMENTS_PURCHASES / PURCHASES
- Cash Dependency Ratio = CASH_ADVANCE / (PURCHASES + 1)
- Payment Discipline Proxy = PRCFULLPAYMENT combined with MINIMUM_PAYMENTS

These features translate raw numbers into interpretable behaviors.

5.2 Feature Selection

Final features should capture: - Spending intensity - Usage frequency - Risk exposure - Credit capacity

Avoid redundant or near-zero variance features.

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6. Choosing the Number of Clusters

6.1 Elbow Method

- Plot inertia vs number of clusters
- Look for diminishing marginal improvement
- Typically suggests 4–5 clusters

6.2 Silhouette Score

- Measures cohesion and separation
- Balance statistical quality with business interpretability

Marketing rule: If a cluster cannot be explained in one sentence, it is not useful.

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7. Model Building

- Fit K-Means on scaled features
- Use multiple initializations for stability
- Fix random state for reproducibility
- Assign cluster labels back to customers

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8. Cluster Profiling and Interpretation

Aggregate key metrics per cluster to understand behavioral differences.

Example Personas

Cluster 1: Low Usage / Dormant Customers - Low purchases and frequency - Low balances **Strategy:** Activation campaigns, introductory offers

Cluster 2: High-Value Responsible Spenders - High spending - High full-payment rate - Low cash advance usage **Strategy:** Credit limit increases, loyalty rewards

Cluster 3: Cash-Advance Dependent Customers - High cash advance frequency - Low full-payment rates - Higher balances **Strategy:** Risk monitoring, financial education, installment offers

Cluster 4: Installment-Focused Planners - High installment purchases - Moderate overall spending **Strategy:** EMI promotions, merchant partnerships

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9. Model Validation

Technical Validation

- Silhouette score

- Cluster size balance
- Stability across multiple runs

Business Validation

- Do clusters align with domain knowledge?
- Are segments actionable?
- Are segment sizes commercially meaningful?

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10. Communicating Results to Stakeholders

Executive Summary

"We identified distinct customer segments with materially different spending, risk, and payment behaviors, enabling targeted marketing and credit strategies."

What Stakeholders Care About

- Size of each segment
- Revenue and risk contribution
- Clear recommended actions

Recommended Visuals

- Cluster size bar chart
- Radar plots per cluster
- Cluster-to-strategy mapping table

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11. Marketing Strategy Mapping

Segment	Behavior	Recommended Strategy
Dormant	Low usage	Activation offers
High-Value	High spend, pays fully	Upsell and loyalty
Risk-Heavy	Cash advance dependent	Risk controls
Installment	EMI focused	Partner promotions

Final Takeaway

This K-Means-based segmentation provides: - Clear behavioral differentiation - Strong business interpretability - Direct linkage to marketing and risk strategies

The framework is scalable, explainable, and suitable for executive decision-making.